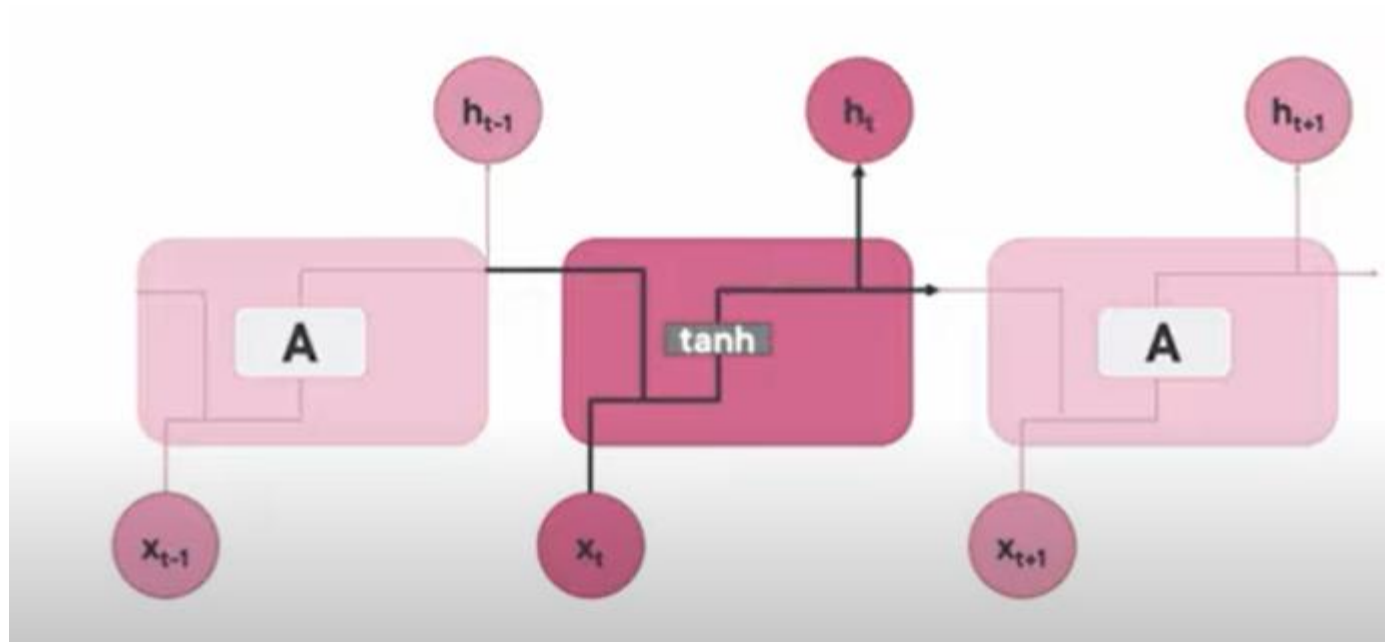
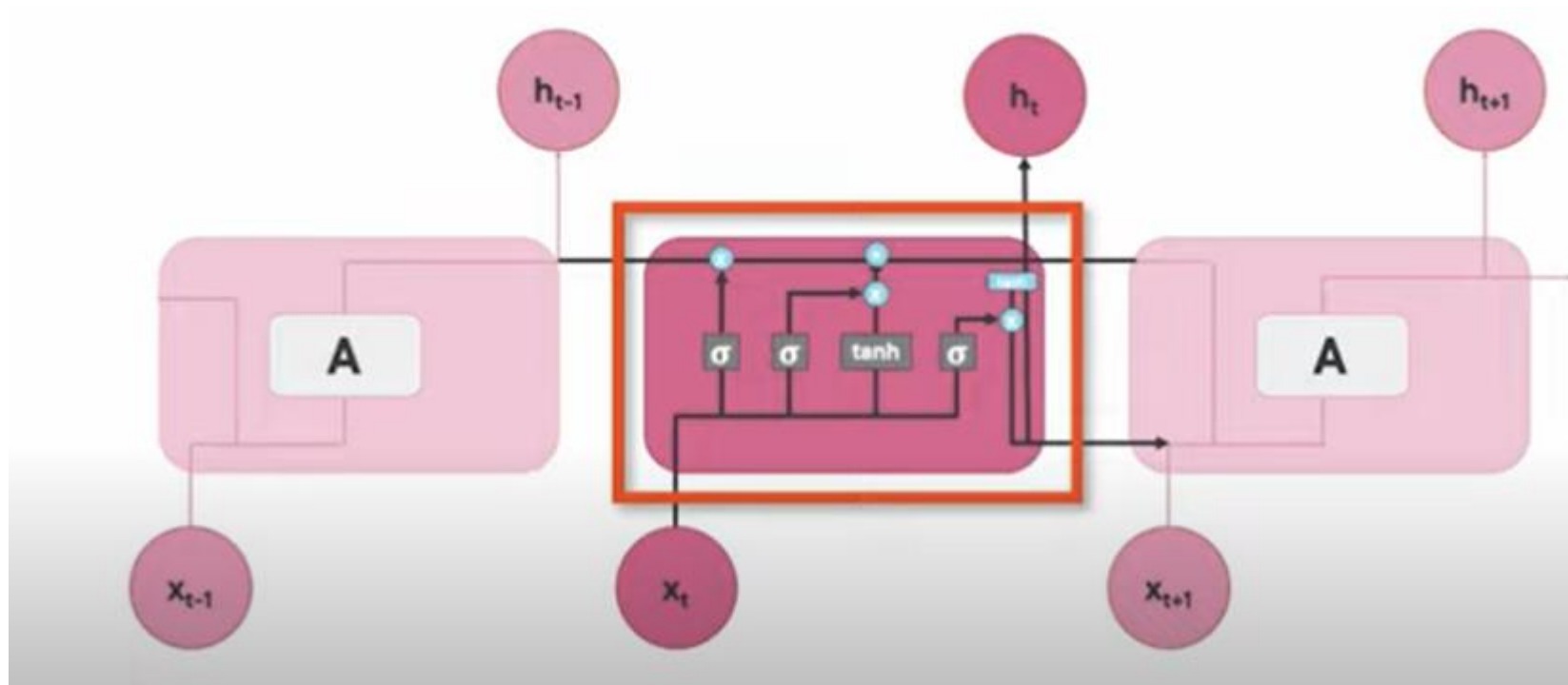


LSTMs



LSTMs

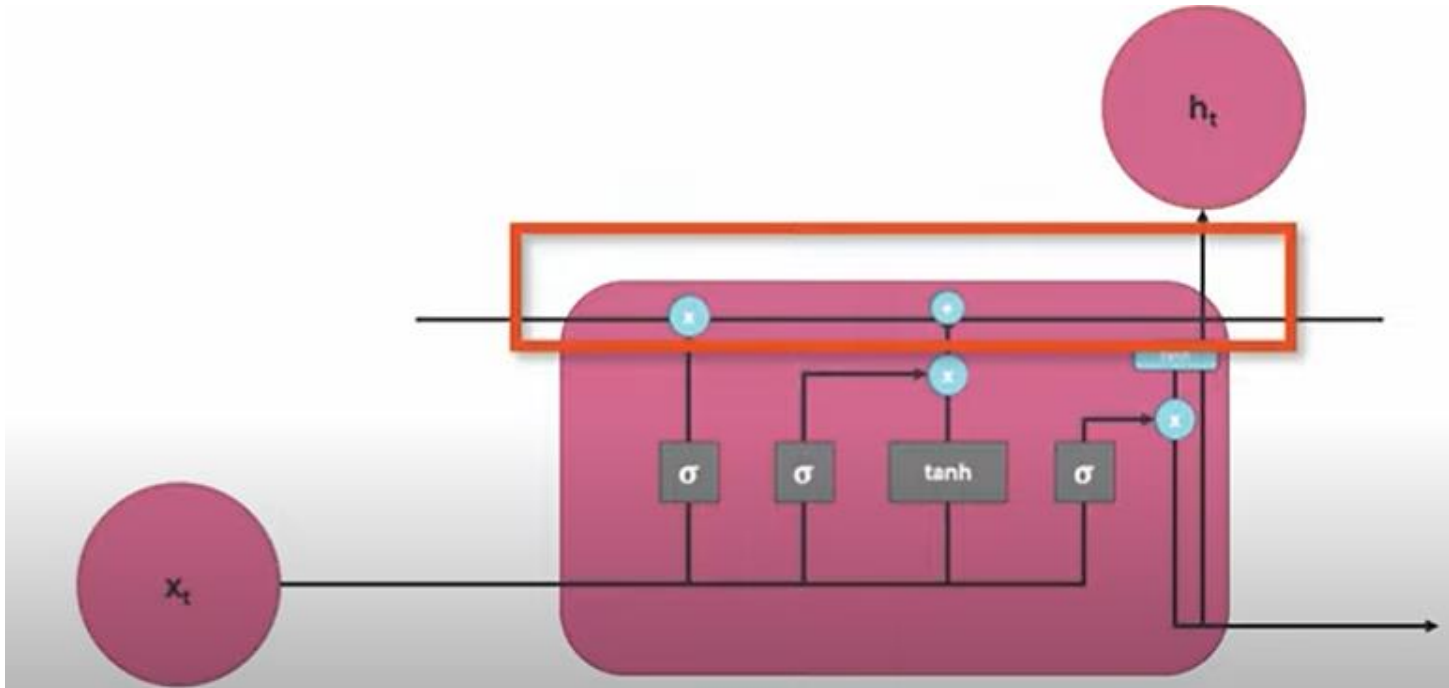


Cell State

Hidden State

Gates (Forget, Input, Output)

LSTMs



The cell state, often denoted as "C," is the primary means by which LSTMs capture and store information over long sequences. It acts as a memory storage unit within the LSTM cell.

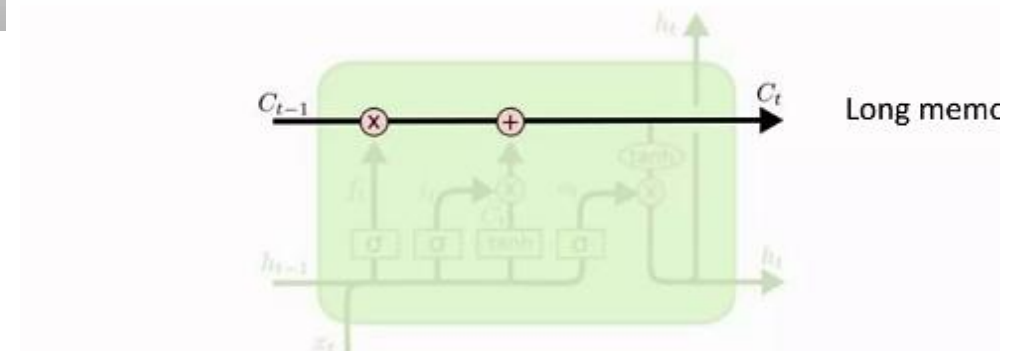
Cell State

Transport highway to transfers information throughout the sequence chain

Information added or removed to the cell state via the gates

Cell state:

- Like a conveyor belt
- It runs straight down the entire chain
- LSTM can remove or add information to the cell state



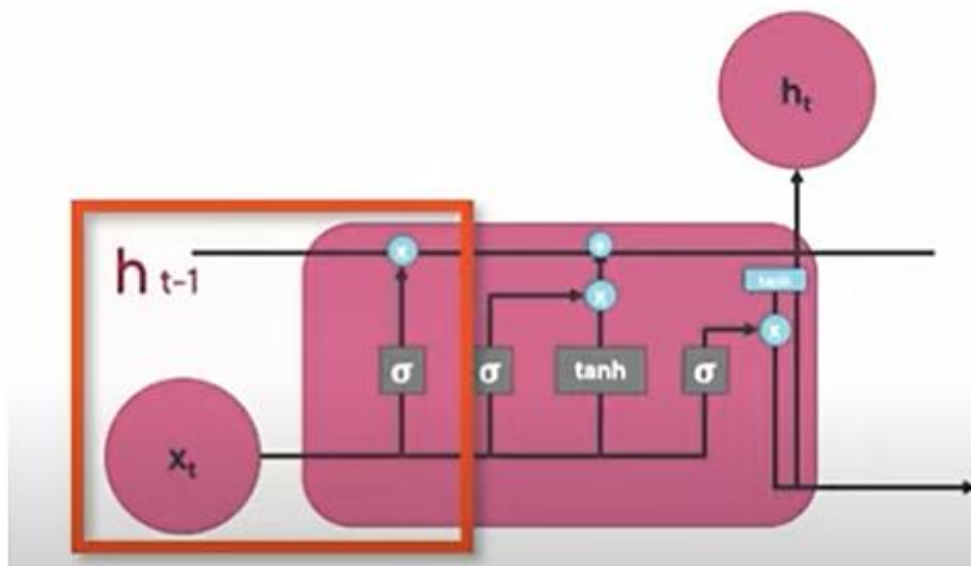
LSTMs

- **Gates:**

- A way to optionally let information through
- They are composed out of:
 - A sigmoid neural net layer
 - A pointwise multiplication operation

- An LSTM has three of these gates, to protect and control the cell state:

- Forget gate layer  Keep gate
- Input gate layer  Write gate
- Output gate layer  Read gate



Gates - Forget

Decide what information to kept or thrown away

Values come out b/w 0 and 1. The closer to 0 means to forget, and closer to 1 means to keep

- **1** represents “**completely keep this**”
- **0** represents “**completely get rid of this.**”

• The hidden state is the output of the LSTM cell at each time step. It is a summary or representation of the information that the LSTM has processed up to that point in the sequence.

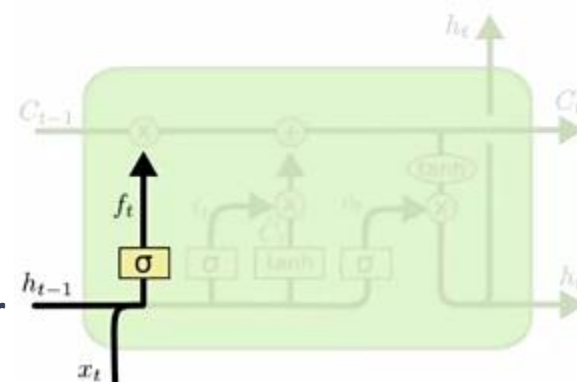
• The hidden state is also influenced by the input data and the cell state but is primarily designed **to capture the short-term or immediate information** needed for the current prediction.

• Forget information:

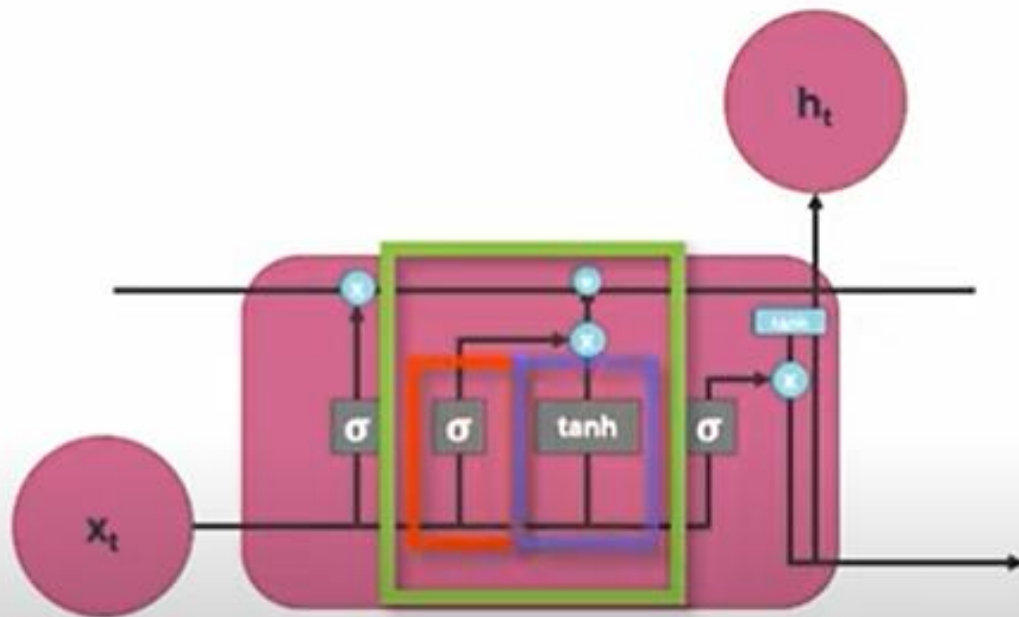
– Decide what information throw away from the cell state

– Forget gate layer:

- Output a number between 0 and 1



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

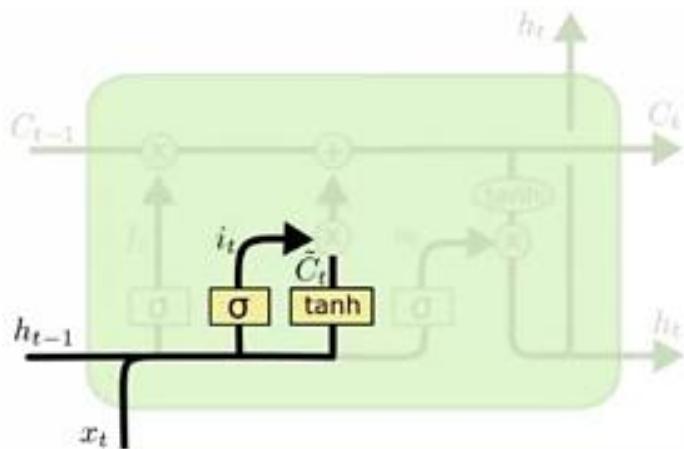


Gates - Input

Has two parts:

- Sigmoid Layer (transform values b/w 0 & 1)
- Tanh Layer (squishing them between -1 and 1)

Sigmoid will decide which information to keep from tanh output



$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

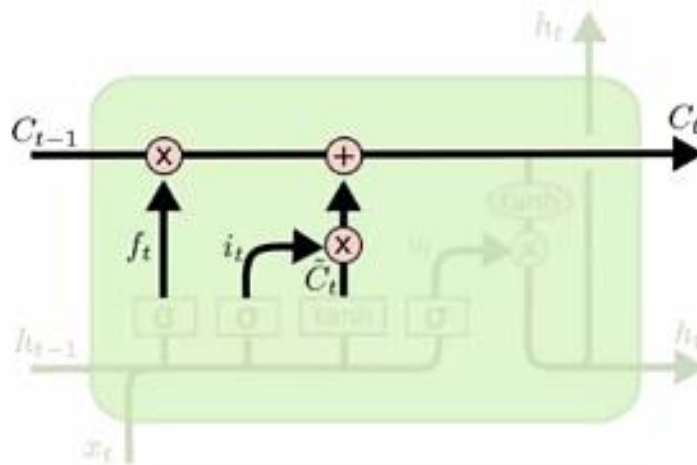
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

• Add new information:

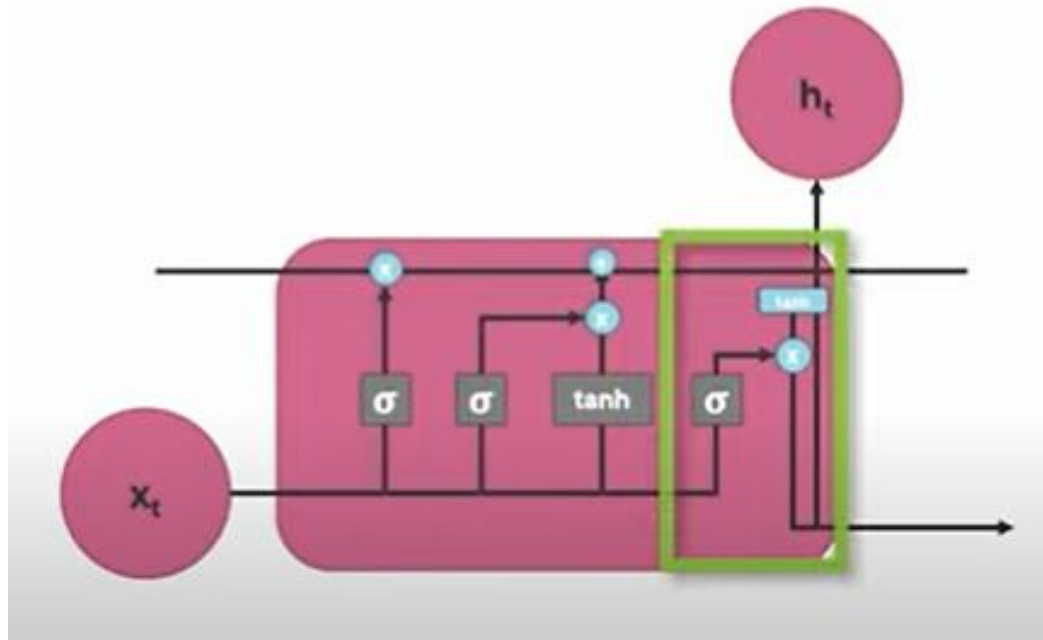
- Decide what new information store in the cell state
- **Input gate layer:**
 - Decides which values we'll update
- **Tanh layer:**
 - creates a vector of new candidate values, $\tilde{C}_t$

- **Update cell state:**

- Forgetting the things we decided to forget earlier: $f_t * C_{t-1}$
- Adding information we decide to add: $i_t * \tilde{C}_t$



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$



Gates - Output

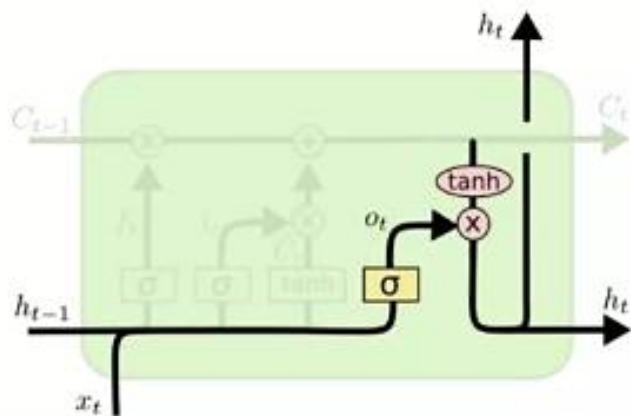
Decide what the next hidden state should be

First, pass information to Sigmoid

Then pass newly modified state to Tanh

Multiply outputs from Sigmoid and Tanh

New hidden state are then carried out to next timestep



$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

Shadow state/ Short memory

- **Create output:**

- Decide what we're going to output

- **Output gate layer:**

- Decides what parts of the cell state we're going to output

- **Tanh layer:**

- Push the values between -1 and +1